

BIALYSTOK, Poland

WMO: 122950

Lat: 53.1073N

Lon: 23.1623E

Elev: 495

StdP: 14.43

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.4	4.8	-5.6	4.2	-0.9	0.4	5.7	6.4	18.2	31.0	16.3	29.2	3.4	50	0.532

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.6	84.0	68.4	80.7	66.2	77.6	64.5	70.2	80.0	68.3	77.4	66.5	74.7	6.3	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.9	100.9	74.6	65.1	94.5	72.5	63.4	89.0	70.4	34.4	80.3	32.9	77.2	31.4	74.7	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.9	13.1	11.7	DB	-6.4	89.2	7.6	2.4	-11.9	90.9	-16.4	92.3	-20.7	93.7	-26.2	95.4	(4)
			WB	-6.9	73.5	7.4	2.0	-12.3	75.0	-16.7	76.2	-20.8	77.3	-26.2	78.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.2	25.6	28.1	33.8	45.4	54.4	60.2	64.3	63.6	54.2	45.1	36.8	29.3	
	DBStd	15.65	10.41	9.86	7.95	7.47	6.95	5.98	5.52	5.56	6.42	7.06	7.80	9.54	
	HDD50	3335	755	613	504	176	34	1	0	0	29	183	398	642	
	HDD65	7386	1220	1033	968	589	332	170	82	93	328	618	846	1107	
	CDD50	1571	0	0	1	38	172	306	444	423	155	30	2	0	
	CDD65	146	0	0	0	0	5	25	61	51	4	0	0	0	
	CDH74	1606	0	0	0	11	97	307	544	594	53	0	0	0	
	CDH80	358	0	0	0	0	6	63	140	142	7	0	0	0	
Wind	WSAvg	5.6	6.3	6.2	6.3	5.8	5.3	5.0	4.8	4.2	4.7	5.6	6.2	6.2	
Precipitation	PrecAvg	24.0	1.3	1.1	1.3	1.5	2.4	2.7	3.3	2.6	2.1	1.9	1.7	1.6	
	PrecMax	36.8	4.3	2.3	3.2	3.3	4.5	5.8	7.4	8.2	6.6	7.9	3.2	3.5	
	PrecMin	17.5	0.2	0.0	0.2	0.2	0.4	0.9	0.2	0.2	0.5	0.1	0.4	0.4	
	PrecStd	4.1	0.8	0.6	0.7	0.7	1.1	1.2	1.7	1.4	1.2	1.4	0.7	0.7	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.2	49.2	59.8	75.4	80.5	86.1	88.6	88.8	80.1	68.2	55.7	48.0	
		MCWB	42.6	44.4	50.1	58.3	63.5	69.0	70.4	69.4	64.8	59.1	52.2	45.8	
	2%	DB	42.6	45.8	53.8	68.0	76.8	81.2	83.8	83.8	74.0	62.2	52.1	45.2	
		MCWB	40.9	42.8	45.7	54.3	61.9	66.6	68.7	68.3	62.6	56.6	49.4	43.3	
	5%	DB	40.3	42.9	48.9	63.3	73.1	77.3	79.8	80.4	69.4	58.4	49.2	42.9	
		MCWB	38.6	40.3	43.5	51.5	59.8	64.3	67.1	65.8	60.2	54.0	47.1	41.4	
	10%	DB	37.6	40.2	45.1	59.6	69.0	73.5	76.1	77.0	65.4	55.8	46.9	40.3	
		MCWB	36.3	38.1	41.1	49.6	57.7	62.2	65.4	63.9	58.1	52.4	45.4	38.9	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.2	45.8	50.8	59.9	66.5	71.1	73.4	72.9	67.0	60.3	53.0	46.5
			MCDB	44.3	47.9	57.4	72.9	76.8	82.3	83.9	83.2	76.6	66.6	55.1	47.3
2%		WB	41.1	43.1	47.3	56.1	63.6	68.5	70.4	69.9	63.6	57.3	49.9	43.9	
		MCDB	42.4	45.3	52.8	65.7	73.6	77.8	79.3	80.4	70.7	60.6	51.6	45.0	
5%		WB	38.7	40.5	44.2	53.5	61.2	65.9	68.5	67.8	61.3	55.0	47.5	41.4	
		MCDB	40.0	42.7	47.8	60.9	70.2	73.7	76.8	77.5	67.8	58.1	48.7	42.7	
10%		WB	36.4	38.0	41.5	50.9	59.0	63.5	66.8	65.7	59.3	52.7	45.5	39.1	
		MCDB	37.5	40.0	44.9	57.0	66.7	70.9	74.4	73.9	64.3	55.3	46.8	40.3	
Mean Daily Temperature Range	5% DB	MDBR	8.4	10.0	13.7	18.8	21.0	19.7	18.6	21.1	17.0	13.6	8.3	7.8	
		MCDBR	8.1	11.2	19.7	26.0	27.1	26.0	24.2	26.9	23.1	18.1	11.4	8.2	
		MCWBR	7.4	9.1	13.2	14.1	13.6	12.8	10.7	12.3	12.5	12.1	9.2	7.1	
	5% WB	MCDBR	7.7	10.7	18.2	23.4	23.5	22.7	21.0	23.4	20.6	16.7	10.2	8.1	
		MCWBR	7.3	9.3	13.0	13.5	12.4	12.1	10.7	11.7	11.7	11.8	8.7	7.4	
Clear-Sky Solar Irradiance	taub	0.304	0.335	0.370	0.417	0.404	0.397	0.431	0.427	0.386	0.358	0.346	0.304		
	taud	2.211	2.198	2.223	2.167	2.269	2.312	2.233	2.240	2.323	2.383	2.278	2.249		
	Ebn at Noon	207	238	255	257	268	271	258	252	248	230	190	185		
	Edh at Noon	21	29	35	42	40	39	41	39	31	24	20	17		
All-Sky Solar Radiation	RadAvg	213	447	842	1290	1667	1750	1641	1417	990	528	204	135		
	RadStd	39	77	125	166	165	148	173	127	144	102	20	22		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
Station Only	N/A	N/A	N/A	N/A	N/A	+1.26	N/A	N/A	N/A	N/A	
Regional (17 neighbors)	+0.83	N/A	N/A	N/A	N/A	+0.99	N/A	-274	+149	N/A	

Nomenclature: See separate page